New Practice Exam

Instructions:

- Read each question carefully.
- Show all your work for full credit.
- You may use a calculator unless otherwise specified.

cky Dice}, 10 points John loves playing games with dice. He has a special 12-sided die, with faces numbered from 1 to 12. Each time John rolls the die, the outcome is uniformly random, independent of other rolls. \{}begin{enumerate}[label=(\{}alph*)] \{}item What is the expected value of the number he rolls? \{}item If John rolls the die 24 times, what is the expected number of times he rolls a number greater than 8? \{}item John rolls the die until he gets a 12. What is the expected number of rolls he makes? \{}item John rolls the die 36 times. What is the expected number of unique numbers he gets? \{}item Suppose John wants to roll a 7. Use the Most Used Inequality, $1-x\{}leq e^{-x}$, to show that if he rolls the die 12 times, the probability that he gets a 7 at least once is at least $1/2$. \{}end{enumerate}

hopping}, 10 points After a long week of work, Sarah decides to go shopping. She has a list of 10 items to buy. Each item has a 0.5 chance of being in stock, independently of other items. \{}begin{enumerate}[label=(\{}alph*)] \{}item What is the expected number of items Sarah will be able to buy? \{}item What is the probability that Sarah will be able to buy all the items on her list? \{}item If each item costs \{}$20, what is the variance of the total amount Sarah will spend? \{}item Sarah will buy herself a treat if she can buy at least 8 items from her list. What is the expected value of the indicator that Sarah gets a treat? \{}end{enumerate}

om Walk}, 10 points Alice decides to take a random walk. She starts at point 0 and with each step, she either moves one step to the right with probability 0.6 or one step to the left with probability 0.4, independently of other steps. \{}begin{enumerate}[label=(\{}alph*)] \{}item What is the expected value of her position after 10 steps? \{}item What is the variance of her position after 10 steps? \{}item What is the probability that Alice returns to the starting point after 10 steps? \{}end{enumerate}

s Errors}, 10 points Michael is a journalist who writes 5 articles every day; for each article he writes, he makes a mistake with probability 0.1 independently of other articles. Rachel is a trainee journalist who writes 3 articles every day; for each article she writes, she makes a mistake with probability 0.2 independently of other articles. \{}begin{enumerate}[label=(\{}alph*)] \{}item Find the standard deviation of the total number of mistakes they make in 7 days. \{}item The two journalists make a total of 3 mistakes on a given day. What is the probability that Michael made more mistakes than Rachel that day? \{}item The next day, at least one of the 8 articles they wrote contains a mistake. Find the expected number of total mistakes they made that day. \{}end{enumerate}

Tickets}, 10 points A lottery has 100 tickets, numbered from 1 to 100. A ticket is randomly picked as the winning ticket. Let \$X\$ be the number on the winning ticket. \{}begin{enumerate} [label=(\{}alph*)] \{}item Define the distribution of \$X\$. What is the expected value of the number on the winning ticket? \{}item What is the probability that the number on the winning ticket is between 50 and 70? \{}item Let \$Y\$ be the indicator random variable that takes the value 1 if the winning ticket has a number between 50 and 70, and 0 otherwise. Find the distribution, parameter, and PDF of \$Y\$. \{}item Suppose John, Sarah, Michael, and Rachel each randomly pick a ticket from the lottery. What is the expected number of times a ticket numbered between 50 and 70 is picked? \{}end{enumerate}